Narra Yaswanth

co24btech11016@iith.ac.in | yashwanthnarra272@gmail.com | +91 8341137485 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.22
Happy valley	Class XII (CBSE)	2024	89.60%
KKR Gowtham	Class X (state board of secondary education AP)	2022	94.10%

PROJECTS

ATSense - Resume Intelligence System *github*

Personal Project **|** Gemini 2.0 Flash, FastAPI, intfloat/e5-small-v2, sentence-transformers, scikit-learn

- Built a **resume parser** supporting **any PDF layout**, including multi-column and designed templates, by reading raw PDF bytes directly and eliminating text extraction failures common in traditional parsing libraries.
- Developed a **multi-resume ATS scoring pipeline** ranking candidates across skills, experience, projects, and education using **intfloat/e5-small-v2 semantic embeddings**, enabling accurate semantic matching over keyword-matching.
- Designed an **automated gap analysis system** that extracts structured candidate and job description data to identify missing skills, weak sections, strengths, and actionable improvement suggestions for every candidate.
- Exposed the complete system through a **FastAPI REST backend**, returning ranked ATS scores and structured comparisons across multiple resumes through a single API interface.

Predictive Maintenance System *github*

Personal Project | Python, pandas, NumPy, scikit-learn, XGBoost, Matplotlib

- Developed an end-to-end machine learning pipeline that predicts equipment failures from multivariate sensor data by performing preprocessing, feature engineering, exploratory data analysis, and feature selection.
- Trained, tuned, and benchmarked multiple machine learning algorithms, selecting the highest-performing model using **Accuracy, Precision, Recall, F1-score, and ROC-AUC** while optimizing predictive performance.
- Visualized feature importance, **confusion matrices**, and model evaluation metrics to improve interpretability, support proactive maintenance planning, and reduce unexpected equipment downtime.

System-Call-Aware Multi-Level Feedback Queue Scheduler (xv6) *github*

Course project | C, xv6, QEMU

- Enhanced the **xv6 operating system** by implementing custom system calls and a **System-Call-Aware Multi-Level Feedback Queue (SC-MLFQ)** scheduler supporting efficient, priority-aware CPU scheduling.
- Modified the kernel to maintain process statistics, track system calls, implement priority-based scheduling, and prevent starvation through a global priority boosting mechanism for fair CPU allocation.
- Developed comprehensive user-space test programs to validate scheduler correctness, analyze scheduling behavior, and evaluate performance across diverse workloads under different scheduling scenarios.

Bank Management System github

Personal Project | Python, OOP, SQLite

- Modeled core banking entities using **OOP principles**, with encapsulated Account, Customer, and Transaction logic.
- Built an **automated transaction monitoring system** applying rule-based checks on amount, frequency, and limits.
- Designed a **custom exception hierarchy** for type-specific error handling, paired with **spending analytics**.
- Persisted account and transaction data using **SQLite**, with validated read and write operations at the class level.

SKILLS

Programming Languages: Python, C, C++

Software Development: FastAPI, SQLAlchemy, REST APIs, JWT, bcrypt

Machine Learning: scikit-learn, XGBoost, TensorFlow, Keras, pandas, NumPy, sentence-transformers, Matplotlib

Databases: MySQL, SQLite

Tools: Git, GitHub, VS Code, Jupyter Notebook, LaTeX

RELEVANT COURSES

Computer Science courses: Introduction to Programming (C), Data Structures and its Applications,

Operating Systems - 1&2

Maths courses: Introduction to Probability, Linear Algebra, Calculus 1&2, Differential Equations, Transform Techniques, Introduction to Statistics, Complex Variables, Numerical Methods

Computational Courses: Fundamentals of Scientific Computing, Computer Aided Numerical Methods - 1&2, Computational Methods in Material Science, Electronic Structure and Atomistic Modeling of Materials

Management courses: Intro to Entrepreneurship

SCHOLASTIC ACHIEVEMENTS

Selected for **Round 2** of **ALGO University**, ranking in the **top 8%** of participants.

Achieved a **LeetCode rating of 1630+**, demonstrating strong problem-solving and DSA skills.

EXTRACURRICULAR

Represented **IIT Hyderabad** as a player in the **Inter IIT Sports Meet 2025 (Hockey)**.

Served as a **Coordinator** for the Hockey event during the **Inter IIT Sports Meet**.

Awarded **Best Player of the Tournament** in the **IIT Hyderabad Hockey League (IHL 2026)**.

Won **Silver** and **Bronze** medals in IIT Hyderabad campus hockey tournaments